ROCK RANGERS: The Great Rock Crew Quest

Ranger Kit | Grade 8 | Team: _____

Meet your guides

IGNA the Lava Lizard

Igneous rocks

"Born in fire, cooled by time!"

Clues: visible crystals, glassy, tiny holes

SEDI the River Turtle

Sedimentary rocks

"Layer by layer, I deliver!"

Clues: layers, gritty grains, fossils, fizzes in acid

META the Mole

Metamorphic rocks

"Heat and pressure, change forever!"

Clues: bands, sparkly crystals, smooth sheets

(Teacher: paste your 3D cartoon character art over these placeholders.)

Team roles

Reader:	reads each clue card aloud
Recorder:	writes on the answer slip
Speaker:	gives the Rock Pitch
Checker:	finds evidence words, runs the app check (Round 3)
Scout (5th member):	helps any role and keeps time

How to play (20 minutes)

ROUND 1 - Clue Hunt (6 min): read the 6 cards, pick a name and a family, circle one evidence word. No answers yet!

ROUND 2 - Rock Pitch (4 min): your Speaker pitches ONE rock to a neighbor team for 45 seconds, then swap.

ROUND 3 - Reveal and Echo (4 min): the answers are revealed. Say the name back, do the guide gesture, and check your score.

FINISH (2 min): add up points and write one sentence on your exit ticket.

Rock Pitch sentence frames

"We believe Rock #__ is _____ because the clue says _____."

"It formed when _____."

"It belongs to _____'s family, the _____ rocks."

ROCK CARD #1

Rock Rangers

Paste rock photo here

The Crystal Mystery

Big, visible crystals in pink, gray and black. Very hard. Cooled slowly deep underground.

Ranger check: ☐ Crystals ☐ Layers ☐ Bands

ROCK CARD #2

Rock Rangers

Paste rock photo here

The Floating Rock

Light and full of tiny holes. Cooled fast from lava. It can float on water!

Ranger check: ☐ Crystals ☐ Layers ☐ Bands

ROCK CARD #3

Rock Rangers

Paste rock photo here

The Beach Rock

Feels gritty like a beach. Made of squeezed sand grains stacked in layers.

Ranger check: ☐ Crystals ☐ Layers ☐ Bands

ROCK CARD #4

Rock Rangers

Paste rock photo here

The Shell Rock

Often has shell bits or fossils inside. Fizzes with weak acid.

Ranger check: ☐ Crystals ☐ Layers ☐ Bands

ROCK CARD #5

Rock Rangers

Paste rock photo here

The Sparkle Rock

Shiny, sparkly crystals and no layers. It was once another rock, then baked and squeezed.

Ranger check: ☐ Crystals ☐ Layers ☐ Bands

ROCK CARD #6

Rock Rangers

Paste rock photo here

The Page Rock

Flat, smooth sheets that split like pages. It was once mud, then squeezed.

Ranger check: ☐ Crystals ☐ Layers ☐ Bands

TEAM ANSWER SLIP

Lock in your answers BEFORE the reveal! Team: _____

ROCK BANK:

Granite Pumice Basalt Sandstone Limestone Shale Marble Slate

Card	Rock name	Family	Evidence word	Pts (teacher)
#1		Igneous / Sedimentary / Metamorphic (circle one)		___ / 3
#2		Igneous / Sedimentary / Metamorphic (circle one)		___ / 3
#3		Igneous / Sedimentary / Metamorphic (circle one)		___ / 3
#4		Igneous / Sedimentary / Metamorphic (circle one)		___ / 3
#5		Igneous / Sedimentary / Metamorphic (circle one)		___ / 3
#6		Igneous / Sedimentary / Metamorphic (circle one)		___ / 3

Round 2 - Rock Pitch notes (Speaker)

Round 3 - App check (Phone Ranger, optional)

App guess: _____ Did it match our answer? Yes / No

If not, who do we trust and why? _____

Exit ticket (each student)

Today I learned that _____ rocks form when _____.

SCORE: Accuracy ___/18 Speaking ___/8 Teamwork ___/4 TOTAL ___/30

Badge: Gold 25-30 | Silver 18-24 | Bronze 10-17 | Trainee <10

TEACHER ONLY - Answer Key and Scoring

Do not print this page for students. Reveal AFTER Round 2.

#	Rock	Family	Guide	Reveal line
1	**Granite**	Igneous (intrusive)	Igna	Slow cooling = big crystals
2	**Pumice**	Igneous (extrusive)	Igna	Fast cooling + gas bubbles = light, can float
3	**Sandstone**	Sedimentary (clastic)	Sedi	Cemented sand grains in layers
4	**Limestone**	Sedimentary (chemical/organic)	Sedi	Shell/fossil material; fizzes in acid
5	**Marble**	Metamorphic (non-foliated)	Meta	Limestone + heat and pressure = interlocking cry stals
6	**Slate**	Metamorphic (foliated)	Meta	Shale/mud + pressure = splits into sheets

Scoring (30 points)

Accuracy (18): per card, 1 pt rock name + 1 pt family + 1 pt valid evidence word (6 cards).

Speaking (8): 0-2 each for clear claim, uses evidence, scientific vocabulary, clear voice/eye contact.

Teamwork (4): 2 pts if every role used; 2 pts if every member spoke at least one sentence.

Bonus (+1): a team that spots an app mistake in Round 3 and explains why.

Echo gestures (Round 3)

Igna: wiggle fingers upward like flames. Sedi: stack your hands layer by layer. Meta: squeeze your fists together.

Printing notes

Print pages 1-3 in color; make 9 sets (8 teams of 4 + 1 team of 5 = 37 students). Cut page 2 along the dashed lines.

Paste real photos from your own samples or Wikimedia Commons into the gray boxes before copying.

Pages are A4. Paper-only teams are never at a disadvantage: the app check in Round 3 is optional.